No Higgs **10** on the $e_6 + a_2$ route of the non-supersymmetric heterotic string

A lattice theorem for $T^6/\mathbb{Z}_3$ orbifolds of the $SO(16) \times SO(16)$ string

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Abstract

In $T^6/\mathbb{Z}_3$ orbifolds of the $SO(16) \times SO(16)$ heterotic string built on a three-family $e_6 + a_2$ shift, Wilson-line breaking to $SU(5)$ leaves fractional charges (Wen–Witten), and the field-theoretic escape needs a scalar in the **10** of $SU(5)$. We show that no such scalar exists for any Wilson-line configuration that leaves an $SU(5)$, in any sector. The proof has two parts: a statistics rule, derived from the projection phase of the fermion-parity element, that decides boson versus fermion from the left-moving momentum alone; and an exhaustive enumeration, by coset tables over $E_8/3E_8$, of the twisted massless **10**-type momenta. Twisted **10**s do occur, in 0.18% of one-line and 1.8% of two-line configurations, but every one has the fermionic value of the rule.

1 Setup

The orbifold group is $\mathbb{Z}_2^W \times \mathbb{Z}_3$. The fermion-parity (Witten) element $g = (-1)^F e^{2\pi i\, p\cdot V_0}$ acts trivially on T^6; its gauge shift V_0 is a *split* involution of $E_8 \times E_8$, $V_0^2 = 2$, the class that makes the ten-dimensional parent the tachyon-free $SO(16) \times SO(16)$ string. The $\mathbb{Z}_3$ shift V_1 is one of seven $e_6 + a_2$ bases with three chiral families; torus one carries a family Wilson line W_f; tori two and three carry order-three Wilson lines W_1, W_2. Sectors are labelled by the Witten label $w \in \{0, 1\}$, the $\mathbb{Z}_3$ power n and fixed-point labels m_a.

2 The statistics rule

Lemma 1. *On a state of sector $h = (w, n, m_a)$ with left momentum P and right-moving weight R, invariance under g requires*

$$P \cdot V_0 - R_0 \;\equiv\; \tfrac{1}{2}\Big(w\,(V_0^2 - 1) + n\,V_1 \cdot V_0 + \textstyle\sum_a m_a\, W_a \cdot V_0\Big) \pmod 1,$$

R_0 the spacetime component of R (integral for bosons, half-integral for fermions). If $V_0^2 = 2$ and all products of V_0 with the shifts and Wilson lines are even, the right side is $w/2$, and a state is a boson iff $P \cdot V_0 \equiv w/2$.

Proof. The phase of g on the state is $(P \cdot V_0 - R \cdot s_0) - \frac{1}{2}(V_h \cdot V_0 - v_h \cdot s_0)$; g is a pure shift, so there are no oscillator or fixed-point phases. With $R \cdot s_0 = R_0$, $v_h \cdot s_0 = w$ and $V_h = wV_0 + nV_1 + \sum m_a W_a$ the formula follows. □

Consistency forces the evenness. If some $W \cdot V_0$ is odd, the right side depends on whether a fixed-point label is written 1 or -2; the projection is ill-defined. The orbifolder rejects every such configuration (147/147), and 65,659 of the 231,124 one-line configurations that satisfy only $W \cdot V_0 \in \mathbb{Z}$ are of this kind. *Check.* From left weights alone the rule reproduces the engine's statistics for all 5,974 fields of eight full $T^6/\mathbb{Z}_3$ spectra and for all 74,174 fields of the 104 A_8 Standard Models of the companion route.

3 The theorem

Theorem 1 (no Higgs **10**)**.** *For every consistent configuration of order-three Wilson lines that leaves an $SU(5)$, no state in the* **10** *of $SU(5)$ is a scalar.*

Proof (computer-assisted). Untwisted: the **10** lies in the **16** of $SO(10)$, with $P \cdot V_0 = \frac{1}{2}$, $w = 0$: fermions (all 7×864 lines). *Twisted:* a massless twisted state has $|P|^2 = 4/3$ (with an oscillator the shell 2/3 is below the **10**-weight norm 6/5); a **10** needs the $SU(5)$ projection of the first-E_8 part to have norm 6/5. Commuting order-three lines diagonalise on the D_5 Cartan, so an $SU(5)$ intersection forces one line of $SU(5)$ type and the other preserving it. For each base, the cosets $V_1 + n_1 W_f + wV_0 + n_2\lambda_1 + n_3\lambda_2$ (and their second-E_8 partners) are fixed by classes in $E_8/3E_8$, so all short vectors are tabulated once and every pair is decided by table look-up. Results:

	consistent configurations	with a twisted **10**	scalar type
one line	165,465	294	0
two lines	119,915,126	2,210,046	0

Every twisted **10**-type momentum of the one-line class lies in a Witten-twisted coset with $P \cdot V_0 \in \mathbb{Z}$; by the lemma all are fermions. Positive controls: an independent random sample of 366 two-line hits is recovered exactly; bases with no hits in the sample have none in the enumeration. □

4 Caveats

- *Wilson-line consistency.* The correct conditions are those of Plöger, Ramos-Sánchez, Ratz and Vaudrevange (hep-th/0702176, eqs. (7a)–(7f)), derived from $\Phi_{\rm vac}(g,h)^{\gcd(n,s)} = 1$ (their (A.11)): in particular $\gcd(N_a, N_b)\,(W_a \cdot W_b) \in 2\mathbb{Z}$ and $N_a(W_a \cdot V_i) \in 2\mathbb{Z}$ — the latter, for V_0, is our parity condition. The non-supersymmetric orbifolder does not impose the first (its failure flag sits in a verbose-only branch). A patched build flags 497 corpus inputs; every one has a lattice-shift repair with a spectrum identical to the stock output (983/983), and at order three the lattice representative carries no torsion (160/160 relabellings invariant). The theorem is a statement of absence and holds in any case.
- *Engine dependence, resolved.* The two $N = 0$ engines agree field for field in the untwisted and $\mathbb{Z}_3$-twisted sectors; the orbifolder 1.2.1-based build drops every Witten-twisted fixed brane (a local condition it applies there fails), which is where all the extra twisted **10**s of the non-supersymmetric engine sit.
- *Scope.* One geometry ($T^6/\mathbb{Z}_3$), one shift class ($e_6 + a_2$), shift orbifolds only.

Consequence. Wilson-line breaking on this route leaves fractional charges, and the field-theoretic escape is closed. Together with the Weinberg-angle and proton obstructions recorded elsewhere, this route does not give a realistic Standard Model.

Reproducibility. Producers `w33_pass12217…`, `12229…`, `12235…`, `12241…`, `12247…`, `12253…` in `analysis/`, with drivers in `analysis/search_drivers_122*`.